

The programs that you run can send packets through virtual switches that seem like real Ethernet interfaces, with a given link speed and delay. Packets get processed by what looks like a real Ethernet switch, router, or middle-box, with a given amount of queuing.

The Mininet CLI and API facilitate easy interaction with our network. Virtual hosts, switches, links and controllers created through Mininet are the real thing. They are just created using the Mininet emulator rather than hardware and for the most part, their behaviour is similar to discrete hardware elements.

How it works

Mininet uses Linux processes in network namespaces on a simple desktop to create scalable, virtual, software defined networks. The installation of the Mininet emulator and its supported package-like controllers (POX, NOX, BEACON and OPENDaylight) and virtual switch (OpenvSwitch) enables us to work with the virtual network environment. Mininet could be run on different platforms and hardware devices with varied capabilities like laptops, servers, virtual machines, Linux systems, or even in the cloud. The default version of Mininet supports some simple predefined network topologies (a network topology is the arrangement of nodes, e.g., hosts, switches and links in a computer network).

If a network programmer needs to create custom topology, Mininet provides the programming support. You can create the custom topology using Python, Java or other supported programming languages. You have to modify existing code or write your own new code to create custom topologies.

Virtualisation of computing resources in every operating system is done through a process of abstraction. The provision of running many hosts and switches on a single OS kernel is done based on process based virtualisation in Mininet, which can build user or kernel-space OpenFlow switches, controllers to control the switches, and hosts to communicate over the simulated network.

Why is it better?

Mininet has various features like emulators, simulators and hardware test-beds that make it superior to other network virtualisation tools.

- Compared to full system virtualisation based approaches, Mininet boots faster.
- It can connect hundreds of hosts and switches through the software network controller.
- It provides higher bandwidth support for network virtualisation on virtual hardware.
- It is inexpensive, easy to install and quickly reconfigurable.
- It supports a variety of virtualisation software.
- It supports Mac, Windows and Linux operating systems with an inbuilt virtual switch and the OpenFlow protocol.

- Easily connects to real networks.

The Mininet installation guide

There are three ways in which Mininet can be installed, and these have been detailed in this section.

Option 1

Mininet VM installation is easy and it's what I would recommend since it's the most foolproof method.

The following steps lead to a VM install:

1. Download the Mininet VM image
2. Download and install a virtualisation system
3. Run through the *VM Setup Notes* to log in to the VM and customise it as desired
4. Follow the *Walkthrough* to get familiar with Mininet's commands and typical usage

Option 2

This is native installation from source, which works well for a local VM, remote EC2, and native installation. It assumes the starting point of a fresh Ubuntu (or, experimentally, Fedora) installation.

For this, I would strongly recommend more recent Ubuntu releases, because they support newer versions of Open vSwitch.

To install natively from source, you first need to get the source code, as follows:

```
git clone git://github.com/mininet/mininet
cd mininet
```

Once you have the source tree, the command to install Mininet is given below:

```
Mininet/util/install.sh [options]
Install.sh option includes
```

- `-a` installs everything that is included in the Mininet VM, including dependencies like Open vSwitch as well as the additions like the OpenFlow Wireshark dissector and POX.
- `-nfv install Mininet` installs the OpenFlow reference switch and OpenvSwitch

Option 3

Installation from packages:

if you're running a recent Ubuntu release, you can install the Mininet packages.

Install the base Mininet package by entering *only one* of the following commands, corresponding to the distribution you are running.

For Mininet 2.1.0 on Ubuntu 14.10:

```
sudo apt-get install mininet
```

For Mininet 2.0.0 on Ubuntu 12.04:

```
sudo apt-get install mininet/precise-backports
```